25900903 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCNBC100 - FUNDAMENTALS OF NUTRITIONAL BIOCHEMISTRY

(2024 ADMISION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. What type of bond involves the sharing of electron pairs between atoms? [K] / [CO1]

   a). Ionic Bond b). Covalent bond

   c). Metallic bond d). Hydrogen bond

2. What characteristic defines essential amino acids? [U] / [CO2]

   a). A) They can be synthesized by the body. b). B) They must be obtained through diet.

   c). C) They need precursors to be synthesised in the body d). D) They can only be found in animal products.

3. Which of the following properties characterizes vitamin C? [K] / [CO3]

   a). A) It is fat-soluble. b). B) It is stored in the body.

c). C) Calcium absorption

d). D) Helps in hormonal regulation.

4. Vitamin E is important for which of the following functions? [U] / [CO3]

a). A) Blood clotting

b). B) Antioxidant defense

c). C) A cyclicic reactions

d). D) Energy production

5. Which of the following is an example of a fibrous protein? [U] / [CO2]

a). Hemoglobin

b). Collagen

c). Myoglobin

d). Insulin

6. What type of bond links amino acids together in a protein? [K] / [CO2]

a). Ionic bond

b). Hydrogen bond

c). Amino acids

d). Disulfide bond

[6x1=6]

Part B

Short answer Questions

Answer any **2** Questions

Each Question carries **2** marks

7. Explain the physiological classification of carbohydrates. [K] / [CO1]

8. Describe the optical isomerism of amino acids and its dependence on pH and side chains. [U] / [CO2]

9. Summarize the classification of fatty acids based on the hydrocarbon chain [U] / [CO2]

10. Describe the role of co-enzyme forms of niacin [K] / [CO3]

[2x2=4]

Part C

Short Essay Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Explain the biological role of vitamin A in our body. [U] / [CO3]

12. Explain the structure of fatty acids, distinguishing between saturated and unsaturated fatty acids. [K] / [CO2]

13. Discuss the role of phospholipids and cholestrol [U] / [CO2]

14. Describe the structure and properties of riboflavin [U] / [CO3]

15. Discuss on the applications of radioactivity [U] / [CO1]

16. Discuss the classification of lipids [U] / [CO2]

[4x5=20]

Part D

Long essay Questions

Answer any **2** Questions

Each Question carries **10** marks

17. Discuss the properties of carbohyrates. [U] / [CO1]

18. Describe the properties and classification of proteins. [K] / [CO2]

19. Describe the structure and properties of pyridoxine [U] / [CO3]

20. Explain the biological function of Folic acid [K] / [CO3]

[2x10=20]